

SCULPTURE MERGE

Game Design Document

INDEX

1. Game Overview.....	3
1.1. Vision Statement.....	3
1.2. High Concept (Elevator Pitch).....	3
2. Core Gameplay.....	3
2.1. Core loop.....	3
2.2. Merge Mechanics.....	4
2.2.1. Piece Tiers.....	4
2.2.2. Merge Rules.....	4
2.3. Control.....	4
2.4. Levels.....	5
3. World Structure & Progression.....	5
3.1. How worlds work.....	5
3.2. World - Chapter - level structure.....	5
3.3. Six worlds.....	5
3.4. World progression.....	6
3.5. Chapter unlock.....	6
4. Level Design.....	6
4.1. Difficulty Curve.....	6
4.2. Level Types.....	7
5. Specials Pieces mechanics.....	7
5.1. Overview.....	7
5.2. Special pieces catalogue.....	7
5.3. Booster.....	8
6. Progression & Meta System.....	8
6.1. Sculpture Collection.....	8
6.2. Star Rating.....	8
7. Economy & Monetization.....	9
7.1. Current System.....	9
7.2. Rewarded Ads.....	9
7.3. IAP tiers.....	9
8. Live Operations.....	9
8.1. Daily System.....	9
8.2. Weekly Events.....	9
8.3. Seasonal Events (Merge Pass).....	10
9. Art Directions.....	10
9.1. Visual Pillars.....	10
9.2. World Visual Theme.....	10
9.3. Key animations.....	11
10. Audio Design.....	11
10.1. Sound Design.....	11
10.2. Music.....	11
HEXA SORT – 30 SECOND VIDEO AD CONCEPT.....	12

1. Game Overview

1.1. Vision Statement

Sculpture Merge is a mobile merge-2 puzzle game in which players combine scrap metal pieces into handcrafted sculptures. Each sculpture is placed directly into a living themed world that fills and the player progresses. The world is the meta and sculpture is the currency to progress.

Sculpture Merge uses sculptures to unlock the next room. Every level produces one sculpture, every sculpture placed unlocks the next level; every four levels completes a world chapter and opens a new themed world.

1.2. High Concept (Elevator Pitch)

Elevator Pitch	Merge scrap into sculptures. Place them into a themed world. Watch the world come to life.
Genre	Merge-2 Puzzle / World-Builder / Casual
Platform	Mobile - iOS 15+ and Android 9+
Target Audience	Casual puzzle players - Ages 16–45
Session Length	5–10 minutes per session, 2–4 levels per session
Monetization	Rewarded ads + IAP (boosters, gem packs, premium world themes)
Tone	Warm, artisanal, optimistic. The satisfaction of making something real with your hands.

2. Core Gameplay

2.1. Core loop

The game's loop is a merge 2 puzzle. The meta loop is a world building progression. The two loops are bound together by the sculpture, the output of every level and the input of every world.

Enter Level	The player taps a level node on the world map. The level opens showing a themed board and a target sculpture silhouette displayed at top-center.
Merge	The board contains 6–14 raw scrap pieces. The player taps a piece to select it, then drags it onto a matching piece of the same tier to merge.
Complete	When the target sculpture is produced, a satisfying assembly animation plays. The sculpture is added to the player's inventory as a collectible object.
Place	The completed sculpture is automatically carried to the world map. The player taps to place it.
World Fills	The world diorama visually updates. The new sculpture appears in the scene, surrounding environment details fill in.
Unlock & Repeat	The next level node on the world map unlocks.

2.2. Merge Mechanics

2.2.1. Piece Tiers

All sculptures are built from a universal tier structure. Every world uses the same mechanics; only the piece art, names, and final sculpture change.

Tier	Generic Name	Viking World Example	Fantasy World Example
T1	Raw Fragment	Iron Shard	Crystal Sliver
T2	Rough Assembly	Riveted Plate	Rune Fragment
T3	Structural Element	Hull Section	Enchanted Beam
T4	Composite Form	Ship Skeleton	Tower Frame
T5	Final Sculpture	Viking Longship	Wizard Tower

2.2.2. Merge Rules

- Only two pieces of the same tier can be merged.
- Two Tier “N” pieces always produce exactly one Tier(N+1) piece. No exceptions.
- Each level's board is pre-defined with enough T1 pieces to reach the target tier via one optimal merge path.
- Merges are permanent, no automatic undo.
- The board has fixed slots. If all slots are filled, no new pieces can appear. Players must merge to free space.
- Special pieces follow the same tier rules but add secondary interactions.

2.3. Control

Tap	Select a piece.
Drag & Drop	Move selected pieces onto a matching piece to merge, or to an empty slot to reposition.
Tap + Hold	Inspect a piece, show tier, name, and what it produces when merged.
Double Tap	Zoom board to fit all pieces in view.
Swipe Down	Exit inspect mode or dismiss tooltip.

2.4. Levels

Every level is defined by four parameters - a target sculpture, a starting board layout, an optimal merge path length, and a difficulty tier.

Parameter	Easy Levels	Hard Levels
Board Size	3×4 (12 slots)	5×5 (25 slots)
Starting Pieces	6–8 T1 pieces	12–14 mixed-tier pieces
Optimal Path Length	3–4 merges	10–14 merges
Special Pieces	None	2–3 active special types

3. World Structure & Progression

3.1. How worlds work

A World is a themed environment that fills up visually as the player completes levels and places sculptures. Each world is divided into four Chapters. Each chapter contains four levels. Completing all four levels in a chapter fills one quadrant of the world.

When all four chapters of a world are complete, the full world is alive and populated. A world complete cutscene plays, the player receives major rewards, and the next world unlocks.

3.2. World - Chapter - level structure

Unit	Count	Description
World	6 worlds	A fully themed environment. Completes when all 16 sculptures are placed.
Chapter	4 per world	One quadrant of the world. Completes when 4 sculptures are placed.
Level	4 per chapter	One merge puzzle. Always produces exactly one sculpture.
Total	96 levels	6 worlds × 4 chapters × 4 levels = 96 base levels.

3.3. Six worlds

- **Viking Age** - Norse coastal village with longhouse and docks.
- **Jungle Temple** - Ancient overgrown temple ruins with waterfall and altar.
- **Deep Sea Outpost** - Underwater research base with coral, submarines, lights.
- **Space Colony** - Off-world habitat on a barren moon with domed structures.
- **Fantasy Kingdom** - Magical floating island with castle ruins and arcane towers
- **Ancient Clockwork** - Massive mechanical city powered by gears and steam

3.4. World progression

Each world diorama is divided into four quadrants. The quadrants are designed so that even a single placed sculpture causes noticeable visual change.

- **Chapter 1** : Ground structures, basic props. The world looks sparse but inhabited.
- **Chapter 2** : Mid-ground structures, functional elements. The world looks operational.
- **Chapter 3** : Distinctive landmarks, atmospheric details. The world has personality.
- **Chapter 4** : Centrepiece sculpture + ambient animations. The world feels alive.

Sculptures are placed in pre-defined slots, the player does not have free placement control. The sculpture rises from the ground, assembles from its component parts, and settles into the scene.

3.5. Chapter unlock

The level map within each world shows four chapters as a vertical progression. Locked levels are visible but greyed out.

Level 1 Completes	Sculpture 1 placed in the world. Level 2 unlocks immediately.
Level 2 Completes	Sculpture 2 placed. Level 3 unlocks. Chapter 1 is 50% complete.
Level 3 Completes	Sculpture 3 placed. Level 4 unlocks. Chapter 1 is 75% complete.
Level 4 Completes	Sculpture 4 placed. Chapter 1 complete. Chapter 2 unlocks.
Chapters 2–4	Same pattern. Each chapter completion triggers a world-progress animation.
World Complete	All 16 sculptures were placed. World complete cutscene. New world unlocks. Major reward granted.

4. Level Design

4.1. Difficulty Curve

Each world's 16 levels follow a consistent difficulty arc. The first two chapters are accessible. The third chapter increases strategic complexity. The fourth chapter is the world's hardest content, designed to feel like a worthy finale before the next world unlocks.

- Chapter 1 - **Easy** - Onboard or re-establish confidence. No special pieces.
- Chapter 2 - **Easy to Medium** - Introduce one special piece type for this world. Board size grows slightly.
- Chapter 3 - **Medium to Hard** - Combine two mechanics. Identifying the optimal path becomes necessary.
- Chapter 4 - **Hard** - Full world mechanics active. Major reward on completion.

4.2. Level Types

- **Main** - Core loop, Single Target sculpture, One optimal merge path, No time pressure.
- **Challenge** - Standard rules with a move count challenge, complete the level in N merges or fewer for a bonus reward. No fail state, players just miss the bonus if they exceed N.
- **Mega Merge** - The world's centrepiece sculpture. Largest board, longest merge path, all world mechanics active.
- **Daily Bonus** - One bonus level per day, outside the main world arc. Produces a cosmetic sculpture variant (golden) for 24 hours.

5. Specials Pieces mechanics

5.1. Overview

Each world introduces one new special piece type alongside standard T1 to T5 pieces. Special pieces follow the same tier rules but have an additional behavior that adds strategic challenge without breaking the core loop.

5.2. Special pieces catalogue

Piece	Introduced	Behavior
Magnetic Piece	World 1	When placed adjacent to a same-tier piece, auto-merges with it on the following turn. Saves one player action.
Locked Piece	World 2	Cannot be merged until a Key Fragment is merged into it. Key Fragments appear on the board independently.
Fragile Piece	World 3	Must be merged within 5 player actions or it shatters into two T1 pieces.
Resonance Piece	World 4	When two Resonance pieces are merged, the result is Tier+(N+2) instead of Tier+(N+1).
Echo Piece	World 5	On merge, leaves a half-strength ghost fragment in its original slot. The ghost can be merged once more, then vanishes. Enables one extra action from a single piece.
Catalyst Piece	World 6	When placed in the centre slot of any 3×3 region, every piece in that region gains +1 tier temporarily.

5.3. Booster

Boosters are consumable items earned through gameplay and purchasable. Using any booster on a level caps that level's challenge-mode rating at 2 stars.

Booster	Effect	Earn / Cost
Hint	Highlights the single next optimal merge.	Daily login, rewarded ad / 20 coins
Undo	Reverses the last merge. Max 3 per level.	Level reward drops / 30 coins each
Board Refresh	Removes all T1 pieces, replacing them with higher-tier fragments.	Rare event reward / 5 gems
Auto-Merge	Performs 2 optimal merges automatically.	Event reward / 2 gems
Extra Slot	Temporarily expands the board by 2 slots for the current level.	Achievement reward / 3 gems

6. Progression & Meta System

6.1. Sculpture Collection

Every sculpture the player builds is permanently added to their Sculpture Collection. A full gallery view accessible from the main menu. The collection displays all 96 sculptures across all 6 worlds, with empty slots for sculptures not yet built.

Each sculpture entry shows stats like sculpture name, world and chapter origin, date first built, total times built.

Rare variant sculptures (golden versions from Daily Forge) have separate collection slots.

Completing a world's full 16-sculpture set awards a World Complete badge displayed in the collection.

6.2. Star Rating

Each level awards 1–3 stars based on merge efficiency. Stars are visible on the world map and feed the chapter unlock visual progress bar.

- 3 Stars - Optimal merge path. No wasted pieces. No boosters used. Completed within the move-count par.
- 2 Stars - Completed with 1–3 extra merges or one booster used.
- 1 Star - Completed with significant waste, multiple boosters, or a board refresh.

7. Economy & Monetization

7.1. Current System

- Coins - Level completion, milestone rewards, daily login, ad rewards.
- Gems - IAP, rare event rewards, 3-star milestones, world complete.

7.2. Rewarded Ads

All rewarded ads are opt-in.

- **Post-level:** Watch ad to double coin reward.
- **Hint request:** Watch ad for one free hint.
- **Board Full warning:** Watch ad to remove one T1 piece.
- **Daily Forge missed:** Watch ad to access yesterday's Forge for 12 hours.
- **Booster trial:** Watch ad to use one Auto-Merge per day at no cost.

7.3. IAP tiers

Pack	Price (USD)	Contents
Starter Pack	Rs 199\-	100 gems, 3× Auto-Merge, 5× Hint
Sculptor's Kit	Rs 499\-	300 gems, 10× Undo, 5× Board Refresh, Bronze Gallery Frame
World Builder Pass	Rs 999\-	750 gems, all boosters ×10, Silver Frame, No Ads 7 days
Master Collection	Rs 1999\-	2000 gems, all boosters ×25, Gold Frame permanent, No Ads 30 days
No Ads Forever	Rs 499\-	Permanent ad removal only. No currency or booster grants.

8. Live Operations

8.1. Daily System

- Daily Reward - 7 day rotating reward calendar. Day 7 always grants a gem bundle. Resets weekly.
- Mega Merge - One unique level per day producing a cosmetic variant sculpture (gold, silver, or rare material finishes). Rewards doubled if completed within 24 hours. Streak of 30 completions awards an exclusive sculpture.

8.2. Weekly Events

World Rush (Mon–Fri)	Complete 8 levels in 5 days. Earns an exclusive limited sculpture that decorates a special showcase slot in the collection.
Forge Festival (Sat)	Double coins on all levels completed that day.
Community Milestone	The entire playerbase collectively completes a target number of sculptures. If the milestone is hit, all active players receive a gem bundle reward.

8.3. Seasonal Events (Merge Pass)

9 seasonal events per year, each lasting 4 weeks. Each event introduces a temporary themed world with 36 unique levels. Event worlds are available only during the event window, then purchasable as a permanent content pack for gems.

Jan	Feb	Mar	April (Break)	May	June	July	Aug (Break)	Sept	Oct	Nov	Dec (Break)
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9. Art Directions

9.1. Visual Pillars

- Metal pieces look heavy, textured, and real. Merge animations convey physical assembly, clangs, and satisfying snap together moments.
- Each world is painted warm. Think hand-crafted model displays. Every object placed in a world looks like it belongs there by design
- Each completed sculpture is the visual climax of its level. It is rendered in higher detail than the board pieces.
- The merge board must be instantly readable. Piece tier is communicated through size, material complexity, and a subtle glow.

9.2. World Visual Theme

World	Palette	Lighting Style
Viking Age	Iron, grey, warm, sea blue	Overcast with firelight warmth
Jungle Temple	Deep green, jade, terracotta	Diffused jungle canopy light
Deep Sea Outpost	Navy, teal, white	Underwater bioluminescence
Space Colony	Dark charcoal, chrome, white	Hard directional moonlight
Fantasy Kingdom	Purple, gold, blue	Magical ambient glow
Clockwork City	Brass, copper, white	Gas-lamp warm industrial

9.3. Key animations

Merge	Pieces slide together, emit a burst, resolve into new pieces. Ease-in-out. Metallic audio
Sculpture Reveal	Board fades, sculpture assembles from its parts centre-screen with a brief hero lighting moment, then transitions to placement screen.
World Placement	Sculpture descends into its world slot with a ground-impact dust. The ambient reacts.
Chapter Complete	The camera pans across the newly filled chapter quadrant. Chapter badge appears with particles.
World Complete	Full cinematic pan across the entire completed world. All sculptures animate simultaneously. Achievement audio.
Fragile Warning	Red pulse outline on fragile pieces, accelerating as countdown nears 1.

10. Audio Design

10.1. Sound Design

Piece Select	Soft metallic tap. Unique tone per piece type.
Merge Success	Two-beat: approach clink + assembly thud. Warm and satisfying.
Merge Failure	Short soft buzz. Non-punishing.
Sculpture Reveal	Rising swell + single brass hit. The level's emotional payoff sound.
World Placement	Ground thud + ambient world sound swell.
World Complete	Full orchestral sting, 6 seconds. The game's biggest audio moment, scaled to the visual reveal.
Booster Use	Unique per booster: Hint = soft bell; Auto-Merge = rapid clink sequence; Board Refresh = whoosh.
Fragile Warning	Creaking/cracking sound that increases in frequency as countdown nears zero.

10.2. Music

- Main menu / World map: Ambient instrumental matching the active world's theme.
- Merge board: Subtle generative underscore. Minimal at level start; builds a layer per major merge tier reached.
- Final levels: A distinct, slightly more intense arrangement that signals the importance of the level.
- Audio settings: Master, music, and SFX independently controllable. Music off by default in silent mode.